Dale Automation

Commissioning - 9.3 Cooler Pump Station, Pipework, Coil & Chiller Setup

Inspection Details

Report #: Dale Automation 2725

Location:

Inspection Date: 29-Jul-2026 04:45 PM

Inspector Name: Sean Frost

Asset Details

Asset Category : Spiral Cooler

Asset Name : Blue Ribbon Aeroton

Asset # : DSS396-9

Meter Reading: 0 Hours

Model/Year: 2024

1. Pump Station & Pipework Layout

SN	Inspection Item	Note	Good	Repair	Replace	NA
1	Pumps	NSCS 80-160/150/P25VCC4 Pump 1 SN: 01624 Motor 1: Lowara 15kW 01311 Pump 2: 01622 Motor 2: Lowara 15kW 01312	✓	○	○	○
2	Isolators	Isolators wired & tested. Logic to switch between pumps & no-run signal if both isolators are switched on verified.	✓	○	○	○
3	Gauges		✓	○	○	○
4	Check valves	Check valves installed as per P&ID. Manual shut-off valves installed either side of strainers. Pneumatic butterfly valves installed on return, piped & wired, logic sequence tested top open/close with pump run signal.	✓	○	○	○
5	Strainers	Strainers installed, cleaned before circulating water.	✓	○	○	○
6	Buffer tank	Buffer tank installed, leak tested. Water treatment supply connected. Pipe connections to tank sufficiently supported. Low-level sensor installed, shut-off protocol verified.	✓	○	○	○
7	Pipework	Pressure tested by BIP, certificate in project file. Pipes cladded sufficiently, valves cladded sufficiently. Cutouts through enclosure roof sealed.	✓	○	○	○
8	Flowmeter	Siemans Sitrans F M Mag 6000I/5100W	✓	○	○	○
9	Bleed Points	Bleed points installed on supply, return & coil stack outlets.	✓	○	○	○
10	Pipe Lagging/Insulation		✓	○	○	○
11	Valve Lagging/Insulation		✓	○	○	○

1. Pumps

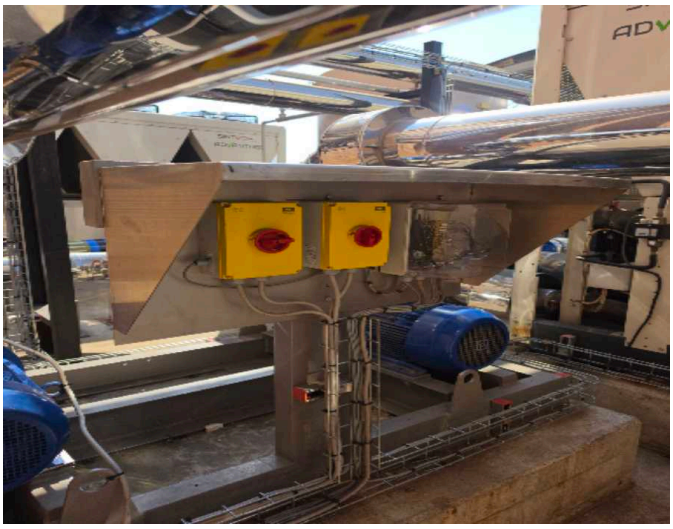








2. Isolators





3. Gauges





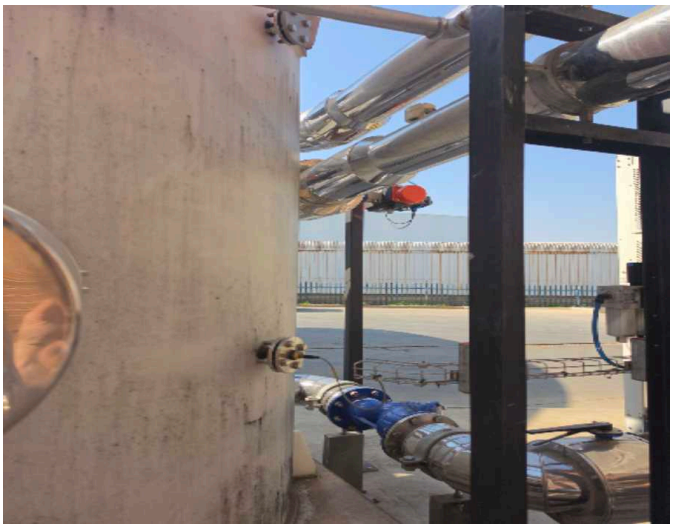
4. Check valves



6. Buffer tank







7. Pipework





8. Flowmeter





2. Chiller Unit

SN	Inspection Item	Note	Good	Repair	Replace	NA
12	Chiller Unit Installation	CGAF 130 SE XLN SN: ELH01622 Damper feet installed as per Trane installation spec. Wired & commissioned by Trane technicians, report in project file.	✓	○	○	○
13	Pump	Grubdfos TPD 100-240/2 X-F-A GQQE-MW1 SN: P22407A Motor 7.5kW 2-pole MG132SB2-38FF265-H3 SN: 8590576010001649	✓	○	○	○
14	Differential pressure switch	Installed & commissioned by Trane technicians. Flow indication displayed on HMI & no-flow pump trip condition verified.	✓	○	○	○
15	Micro strainer	Micro strainer installed & cleaned prior to pump commissioning.	✓	○	○	○
16	System performance	Design setpoint: 10°C Actual SP: 7°C for cooler setpoints 14°C UD, 10°C DD	✓	○	○	○

SN	Inspection Item	Note	Yes	No	NA
17	Chiller Unit Commissioning	Evaporator flowrate: 26l/s @0.8Bar	✓	○	○

12. Chiller Unit Installation

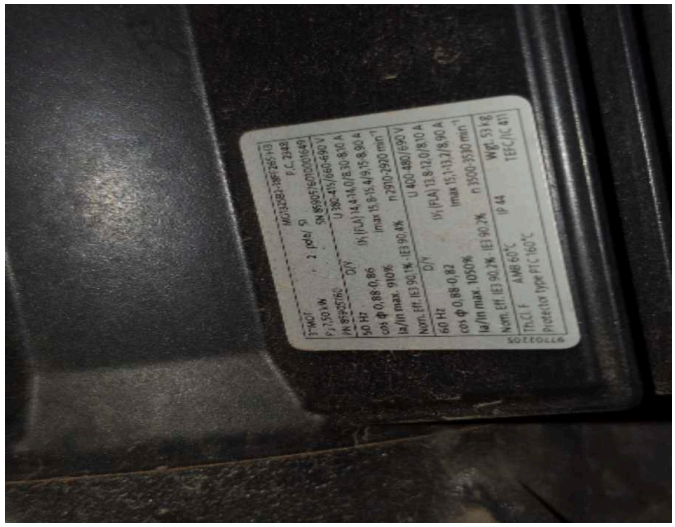






13. Pump





14. Differential pressure switch





15. Micro strainer



17. Chiller Unit Commissioning





4. Coil Balancing

SN	Inspection Item	Note	Good	Repair	Replace	NA
18	Up Drum Bottom Stack	Coil & bypass flowrate set @8.2l/s.	✓	○	○	○
19	Up Drum Middle Stack	Coil & bypass flowrate set @8.2l/s.	✓	○	○	○
20	Up Drum Top Stack	Coil & bypass flowrate set @8.2l/s.	✓	○	○	○
21	Down Drum Bottom Stack	Coil & bypass flowrate set @3.6 l/s.	✓	○	○	○
22	Down Drum Middle Stack	Coil & bypass flowrate set @3.6 l/s.	✓	○	○	○
23	Down Drum Top Stack	Coil & bypass flowrate set @3.6 l/s.	✓	○	○	○
24	Up Drum Coil Face Velocities	Averaging 2.3m/s face velocity @45Hz SP.	✓	○	○	○
25	Down Drum Coil Face Velocities	Averaging 2.1-2.2 m/s face velocity @45Hz SP.	✓	○	○	○

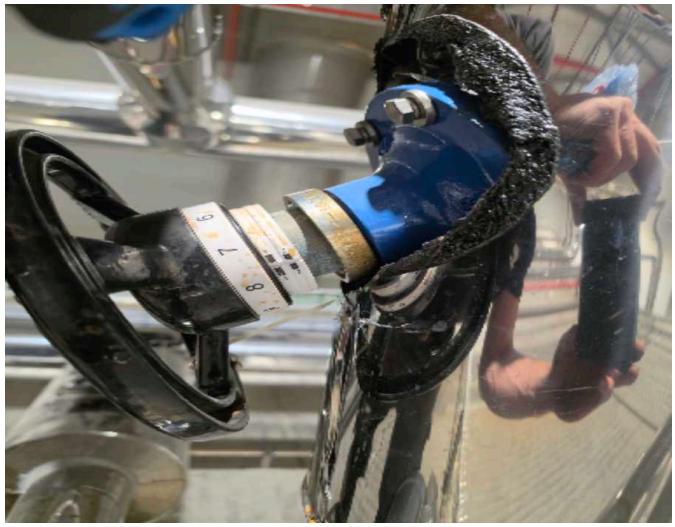
26	Circulation Pump Performance	2.8-3Bar supply pressure @full load. Flowrate SP: 38l/s @47Hz Power draw: 12.9kW Current: 22.8 A	✓	○	○	○
27	Up Drum Fan Performance	Frequency: 45 Hz Current: 6.5-7A Power: 2.9-3.4 kW	✓	○	○	○
28	Down Drum Fan Performance	Frequency: 45 Hz Current: 6.5-7A Power: 2.9-3.4 kW	✓	○	○	○

18. Up Drum Bottom Stack





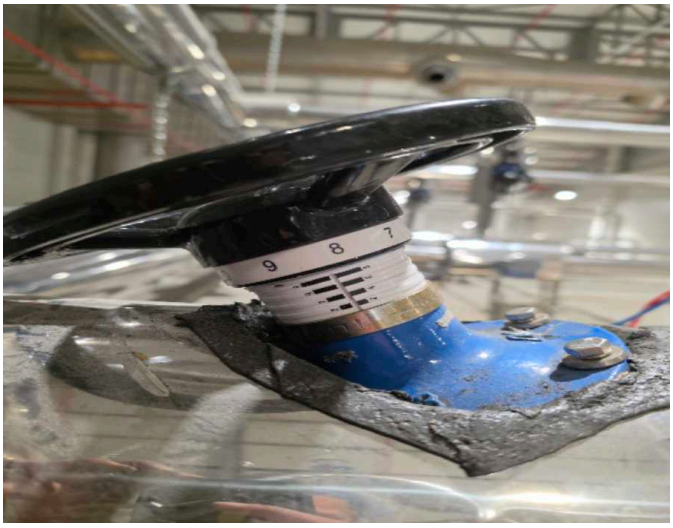




19. Up Drum Middle Stack

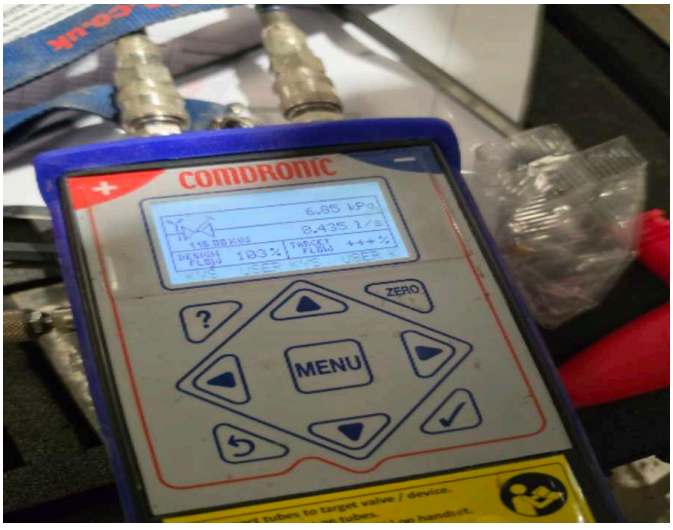


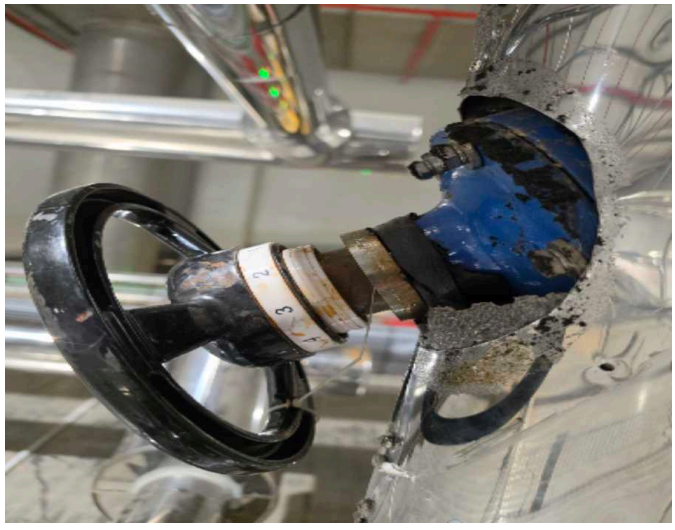






20. Up Drum Top Stack





21. Down Drum Bottom Stack









22. Down Drum Middle Stack









23. Down Drum Top Stack









24. Up Drum Coil Face Velocities







25. Down Drum Coil Face Velocities







26. Circulation Pump Performance









Additional Note

Overall Condition:	Excellent Condition
Asset Safe to Use:	**Yes**
Maintenance Required:	No
Asset Status	Available

Custom Business Fields

Client Name : Blue Ribbon Aeroton Plant 2

Contact details : 396-9



Dale Technician Signature

Client Signature